

WATCH EVERYTHING. TOUCH NOTHING.

Threat intelligence from traffic you can **only watch**.

Critical networks are monitored through **one-way taps**: you see every packet but can never reply, probe, or block. We turn that read-only stream into **live, explained alerts**, and we flag malware hiding **inside encrypted sessions** from its fingerprint alone, no decryption.

▶ 2-MIN LIVE DEMO



Scan to watch it run

youtu.be/your-demo-id

IDEA TITLE

Passive Streaming Threat Detector

EDITION

Software · Offline / Air-gapped

TEAM

[Team Name] · [Team ID]

Six threats, six passive signals, one live verdict

THE REAL SETTING

Power grids and defence links are watched through a **data diode**: traffic is copied one way so the monitor can never become a way in.

WHY IT'S HARD

No probing, no handshakes, **no decryption, no blocking**. Most tools assume two-way access. Passive-only, streaming, multi-threat detection is rare.

OUR ANSWER

A **streaming pipeline** where each threat gets its own detector, and **every alert carries the evidence** that fired it.

WHAT IT DETECTS, FROM METADATA ALONE

- **Floods, beacons, scans, tunnels, exfiltration** — each read from a different statistical fingerprint, not a payload.
- Alerts stream in with **bounded latency** and a standard schema: time, flow, threat class, confidence, evidence.
- Runs entirely **offline** on a read-only feed. Nothing is ever sent back across the tap.

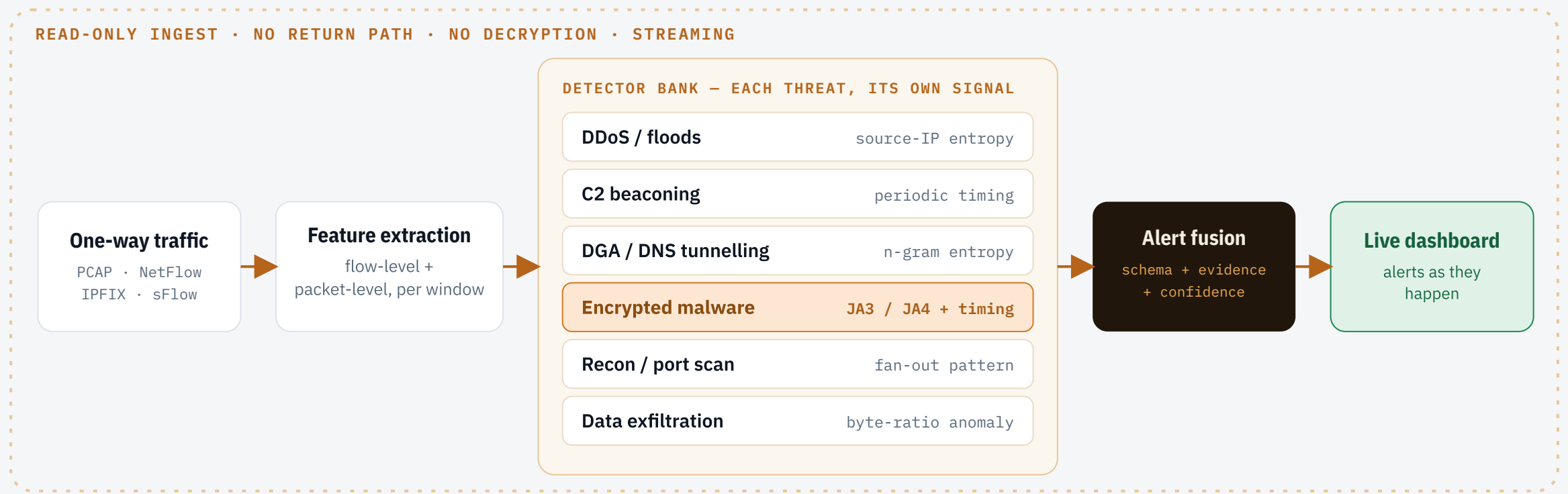
INNOVATION · WHAT NO ONE ELSE BUILDS

Malware caught inside encrypted traffic

Everyone inspects payloads. We can't, and don't need to. Using **JA3 / JA4 TLS fingerprints** plus packet-size and timing patterns, we flag **malware inside encrypted TLS/QUIC sessions without decrypting a byte** — the hardest ask in the brief, and the one most teams skip.

A streaming pipeline built for a one-way world

Read-only in, evidence-backed alerts out. Each detector reads the signal that its threat cannot hide.



The data is public, the models are light, the demo is honest

WHY IT IS FEASIBLE

- **Labelled data already exists.** CIC-IDS-2017/2018 and CTU-13 ship attack timelines, so every detector has ground truth to train and prove against.
- **No GPU needed.** Per-threat models are lightweight (entropy stats, tree ensembles), fast enough to run streaming on a laptop.
- **Streaming is standard.** Windowed processing with per-flow state, then replay a capture at a controlled rate to show it live.

Grand-finale MVP: **a live dashboard replaying real traffic, 4 detectors including the encrypted-malware one, a stated throughput, and F1 measured against a simple baseline.**

RISKS, AND HOW WE BEAT THEM

"Is it really live?"

We replay a real capture at a set rate and show **bounded per-alert latency** and sustained flows/sec, not an end-of-run summary.

Encrypted-malware accuracy

We pair a **known-bad JA3/JA4 list** with a trained metadata classifier and **report honest precision/recall**. We claim detection support, never certainty.

False positives on a monitor that can't block

Alerts are **ranked by confidence with evidence attached**, so analysts triage the top of the list. We measure FP rate against a baseline.

It sees the attack forming, without ever becoming part of the attack surface

Before compromise

Flags recon and beaconing as they unfold, not in a post-mortem after the breach.

Zero added exposure

Read-only by design. The detector can never be pivoted into the protected network.

Encrypted ✓ visible

Sees threats inside TLS/QUIC from metadata, so privacy and chain-of-custody stay intact.

WHY THIS MATTERS

- **Critical infrastructure is watched passively for a reason.** This makes that safe, one-way vantage genuinely intelligent instead of just a recorder.
- **Encrypted traffic is now the norm.** Detecting threats without decryption keeps monitoring lawful, private and evidentially clean.
- **Explainable alerts fight fatigue.** Every alert names the feature that fired, so analysts act on signal instead of drowning in noise.

WHO IT SERVES

NCIIPC & CERT-In
National critical-infra monitoring

Power / telecom / banking
One-way tap deployments

Defence enclaves
Air-gapped monitoring networks

SOC teams
Evidence-backed, triage-ready alerts

National angle: indigenous, air-gapped threat intelligence for India's critical networks, protecting them **without adding a single new way in.**

Data, techniques and evidence behind the build

DATASETS	CIC-IDS-2017 / 2018 (Canadian Institute for Cybersecurity) · CTU-13 botnet dataset — labelled attacks with timelines.
ENCRYPTED-TRAFFIC	JA3 / JA4 TLS client fingerprinting (Salesforce / FoxIO) — detection without decryption.
THREAT MAPPING	MITRE ATT&CK stages · NetFlow / IPFIX / sFlow flow records
TOOLING	Scapy, PyShark, scikit-learn — parsing, features, lightweight models.
CONCEPT	Data diode / unidirectional gateway — the read-only monitoring model we design for.
PROBLEM	SIH26145 · National Technical Research Organisation (NTRO)

ON HONEST EVALUATION

We benchmark every detector's **precision, recall and false-positive rate** against a simple baseline on the same public data, and state the **throughput** we sustained. Numbers shown at the finale come from our own runs, not the brief.

Note on data: no live intercept or sensitive traffic is required. All training and demo use open, published datasets.